DHARANEESH M

Hardware Engineer · Embedded Systems · IoT & Wireless Communication
+91 7708222195 · Chennai, India · github.io/dharaneeshm · **MIT Anna University · ECE**

PROFILE

Hardware engineer in the making — and already competing at the national level.
Proven record of leading teams to finals across national-level competitions, consistently placing among the top selections out of tens of thousands of entries. ECE student at Madras Institute of Technology, Anna University — hands-on across embedded systems, PCB design LoRa wireless, IoT, and bench instrumentation. Not just a participant — a finisher who delivers under pressure, every time.

TECHNICAL SKILLS

Microcontrollers: ESP32, Arduino Uno, Arduino Nano — bare-metal firmware, ADC, GPIO, peripheral integration
PCB Design: KiCad — single & multi-layer, schematic capture, trace routing, ground planes, Gerber output, DFM
Protocols: I2C, UART, LoRa 868 MHz (mesh topology), GPS interfacing, IoT integration
Simulation: LTspice (SPICE), QGIS (geospatial / satellite analysis), AutoCAD, SolidWorks
Programming: C — Intermediate · C++ — Intermediate · Python — Basic

PROJECTS

ESP32 Digital Oscilloscope · *Prototype Complete*

- **Problem:** Every student lab depends on expensive bench oscilloscopes — inaccessible for personal learning, field debugging, or low-budget institutions.
- **Solution:** Built a fully functional oscilloscope from scratch on ESP32 with real-time ADC waveform capture and live display on SSD1306 OLED over I2C — no pre-built modules, no shortcuts.
- **Engineering depth:** Tuned Vmax, Vmin, sampling rate, and frequency range to student-lab spec; validated every output against a bench DSO using function generator references. Full bring-up: schematic → breadboard → firmware → debug.
- **Advantage:** A pocket-sized, low-cost measurement tool that puts signal analysis capability in the hands of any engineer, anywhere.

Impact: *Proves end-to-end embedded hardware ownership — from analog signal to pixel on screen.*

LoRa Emergency Mesh Communication Network · *Smart India Hackathon — College Selected · Prototype*

- **Problem:** In floods, earthquakes, and rural emergencies, cellular networks collapse exactly when people need them most — leaving rescue teams and victims with no way to communicate.
- **Solution:** Engineered a decentralised 868 MHz LoRa mesh network achieving ~2000 m campus coverage with zero cellular, Wi-Fi, or internet dependency — self-healing, self-routing, deployable in minutes.
- **Engineering depth:** Each node functions as both receiver and relay; integrated GPS for real-time coordinate broadcast; custom power circuit for field durability. Stack: Arduino + LoRa transceiver + GPS.
- **Advantage:** Coverage scales linearly — add one node, extend the network. Infrastructure cost per node is a fraction of any alternative.

Impact: *Selected at Smart India Hackathon college level — one of India's most competitive engineering platforms.*

UZHAV — Flood Early Warning System for Farmers · *Tenkasi Hackathon Runner · Best Paper ×2 · Ideation*

- **Problem:** Farmers in flood-prone rural India have no reliable early warning system — they have no smartphones, no GSM signal, and no time when floodwater rises.
- **Solution:** Architected UZHAV: QGIS satellite imagery processes flood zone risk in real time → data pushed over 868 MHz LoRa mesh → alerts broadcast in the farmer's native language to low-cost receiver nodes on their land. Zero internet. Zero smartphone required.
- **Engineering depth:** End-to-end offline pipeline — satellite → computation → radio → human alert. Designed for last-mile deployment in the most infrastructure-poor environments in India.
- **Advantage:** A farmer gets a flood warning in their own language, on a device that costs less than a meal — hours before the water arrives.

Impact: *Best Business Paper — MIT Anna University · Best Technical Paper — eFocus Symposium, MIT ECE · Tenkasi Hackathon Runner.*

CureVox — Medical Emergency Alert System · *Mumbai Hackathon — Top 3,000 / 20,000+ · Ideation*

- **Problem:** In India, the average emergency response time costs lives — patients reach hospitals too late, with no prior medical context for the treating team.
- **Solution:** Led hardware architecture for an autonomous IoT alert node: ESP32 + vitals sensors (pulse, temperature, SpO2) + GSM + GPS — fires patient data and location to hospitals at the moment of emergency, before the patient even arrives.
- **Advantage:** Hospitals receive live vitals + location simultaneously — enabling pre-prepared treatment and faster intervention. Scalable to any rural clinic or ambulance.

Impact: *Top 15% of 20,000+ national entries — 12-member team targeting MSME and government adoption across India.*

PCB Design Portfolio · *KiCad · Single & Multi-Layer*

- **Scope:** BJT amplifier boards (CE/CB/CC), clipper and clamper circuits — designed from first principles, not templated.
- **Execution:** Trace width calculated from current requirements, ground planes poured for noise immunity, DFM-reviewed for manufacturability, Gerber packages generated for fabrication.
- **Advantage:** Demonstrates the ability to take an analog circuit from hand-calculation to a board that a factory can build — a skill most engineers only develop years into a job.

EXPERIENCE

Hardware Engineering Intern · *National Institute of Ocean Technology (NIOT) — Govt. of India, Ministry of Earth Sciences*
May 2026 – Present

- **Context:** NIOT is India's premier ocean technology R&D institute — embedded systems here operate in marine environments where failure is not an option.
- **Role:** Assigned to image processing and embedded instrumentation division — signal interfacing, sensor acquisition pipelines, and hardware validation for oceanographic field systems.
- **Standard:** Working under defence-grade engineering discipline: precision documentation, rigorous fault-tolerance protocols, and system-level testing that academic labs never demand.

Impact: *Government-validated hardware engineering experience — institutional credibility that no coursework can replicate.*

EDUCATION

B.E. Electronics & Communication Engineering · *Madras Institute of Technology, Anna University — Chennai* 2024 – Present

- CGPA: 8.3 / 10.0 · Coursework: Embedded Systems, Analog & Digital Electronics, Signal Processing, Wireless Communication, Microprocessors, IoT, Control Systems.

ACHIEVEMENTS & RECOGNITION

- **Best Business Paper —** MIT Anna University (2024–25): UZHAV flood early warning system.
- **Best Technical Paper —** eFocus Symposium, Dept. of ECE, MIT: UZHAV LoRa wireless architecture.
- **Binary Blitz — 1st Place:** Digital Electronics MCQ, Ascentron Symposium, MIT ECE.
- **Tenkasi Hackathon — Runner-up:** UZHAV validated by industry and government panel judges.
- **Mumbai Hackathon — Top 3,000 / 20,000+:** CureVox medical alert system, national top 15%.
- **Smart India Hackathon — College Level Selected:** LoRa emergency communication prototype.
- **NIOT Internship:** Competitive placement at Govt. of India R&D institute, Ministry of Earth Sciences.

ADDITIONAL

Certifications: UI/UX Design Fundamentals — Google · Entrepreneurship — Google
Languages: English (Professional) · Tamil (Native)
Focus Areas: Disaster-Tech Hardware · Wireless Embedded Systems · PCB Design & Fabrication · Smart Healthcare IoT